Practical 1: Setting Up Your Development Environment

Day 1 — Duration: 3 Hours

SDI Training for ICT Professionals — Climate Risk Database

What you will accomplish: By the end of this practical you will have Docker Desktop installed and running, QGIS installed and ready for mapping, verified access to the live CRD platform, and practised 10 essential Linux/terminal commands. No previous programming experience is needed.

Prerequisites Checklist

Before you begin, please confirm you have the following:

- ☐ A laptop with at least 8 GB of RAM and 20 GB of free disk space
- ☐ Administrator (admin) rights on your laptop — you need these to install software
- ☐ A working internet connection
- ☐ Your operating system: Windows 10/11, macOS 11+, or Ubuntu 22.04+
- ☐ A web browser (Chrome or Firefox recommended)

Important: If you do not have administrator rights, ask your IT department before the training starts. Installation will fail without admin access.

Part 1: Install Docker Desktop

Docker is a tool that lets you run applications inside isolated containers. Think of a container as a lightweight virtual computer that runs a single application. We will use Docker to run the CRD platform on your own laptop.

Windows Installation

Step 1: Open your web browser and go to: `https://www.docker.com/products/docker-desktop/`

Step 2: Click the blue **"Download for Windows"** button. A file called `Docker Desktop Installer.exe` will download.

Step 3: Double-click the downloaded file. If Windows asks "Do you want to allow this app to make changes?" click **Yes**.

Step 4: Follow the installer screens — accept all the default options. When it asks about WSL 2 (Windows Subsystem for Linux), make sure the box is checked.

Step 5: When installation finishes, click **"Close and restart"** to restart your computer.

Step 6: After your computer restarts, Docker Desktop should start automatically. You will see a whale icon in the bottom-right system tray. Wait until it says **"Docker Desktop is running"**.

macOS Installation

Step 1: Open your web browser and go to: `https://www.docker.com/products/docker-desktop/`

Step 2: Click **"Download for Mac"**. Choose **Apple Silicon** if you have an M1/M2/M3 Mac, or **Intel** if you have an older Mac. (Not sure? Click the Apple menu → About This Mac → look for "Chip".)

Step 3: Open the downloaded `.dmg` file. Drag the Docker icon into the Applications folder.

Step 4: Open **Applications** and double-click **Docker**. If macOS asks for permission, click **Open** and enter your password.

Step 5: Wait for Docker to start. You will see a whale icon in the top menu bar. When the animation stops, Docker is ready.

Ubuntu Installation

Open a terminal window (press `Ctrl+Alt+T`) and run these commands one at a time:

Step 1: Update your package list:

```
sudo apt update
```

What this does: Downloads the latest list of available software from Ubuntu's servers so your system knows what can be installed.

You should see: Several lines of text ending with "Reading package lists... Done".

Step 2: Install required helper packages:

```
sudo apt install -y ca-certificates curl gnupg lsb-release
```

What this does: Installs small utilities that Docker's installer needs to work properly.

You should see: Lines showing packages being downloaded and installed, ending with "done".

Step 3: Add Docker's official download key:

```
sudo install -m 0755 -d /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | sudo gpg --dearmor -o /etc
    /apt/keyrings/docker.gpg
sudo chmod a+r /etc/apt/keyrings/docker.gpg
```

What this does: Adds a security key so Ubuntu trusts software downloaded from Docker's servers.

You should see: No output (silence means success for these commands).

Step 4: Add the Docker software source:

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]
     https://download.docker.com/linux/ubuntu $(lsb_release -cs) stable" | sudo tee /
    etc/apt/sources.list.d/docker.list > /dev/null
```

What this does: Tells Ubuntu where to find Docker software for downloading.

You should see: No output.

Step 5: Install Docker:

```
sudo apt update && sudo apt install -y docker-ce docker-ce-cli containerd.io docker-
    compose-plugin
```

What this does: Downloads and installs Docker and all its components.

You should see: Many lines of download progress, ending with "done".

Step 6: Allow your user to run Docker without typing `sudo` every time:

```
sudo usermod -aG docker $USER
```

What this does: Adds your user account to the "docker" group, giving you permission to use Docker directly.

You should see: No output.

Step 7: Log out and log back in (or restart your computer) for the group change to take effect.

Verify Docker Installation (All Operating Systems)

Open a terminal (Windows: search for "PowerShell"; Mac: open "Terminal" from Applications → Utilities; Ubuntu: press `Ctrl+Alt+T`).

Step 1: Run the Docker test:

```
docker run hello-world
```

What this does: Downloads a tiny test container and runs it. This proves Docker is installed and working correctly.

You should see:

```
Hello from Docker!
This message shows that your installation appears to be
working correctly.
...
```

Step 2: Check the Docker version:

```
docker --version
```

What this does: Shows which version of Docker you have installed.

You should see: Something like `Docker version 27.x.x, build xxxxxxx`

If "docker run hello-world" fails:

- **Windows:** Make sure Docker Desktop is running (whale icon in system tray). If you see a WSL error, open PowerShell as Administrator and run: `wsl -update`
- **Mac:** Make sure Docker is running (whale icon in the top menu bar).
- **Ubuntu:** Did you log out and back in after Step 6? Try that first. If it still fails, try: `sudo docker run hello-world`

Part 2: Install QGIS

QGIS is a free, professional Geographic Information System (GIS) application. We will use it to view maps, style geographic data, and create professional map outputs.

Download and Install

Step 1: Open your web browser and go to: `https://qgis.org/download/`

Step 2: Download the installer for your operating system:

- **Windows:** Click the **Long Term Release (LTR)** button for Windows. Download and run the `.msi` installer. Accept all defaults.
- **Mac:** Click the **Long Term Release (LTR)** button for macOS. Download the `.dmg` file, open it, and drag QGIS into Applications.

- **Ubuntu:** Run the following commands in a terminal:

```
sudo apt install -y qgis qgis-plugin-grass
```

What this does: Installs QGIS and an additional plugin for advanced spatial analysis.

You should see: Download progress followed by "Setting up qgis... done".

Step 3: Open QGIS:

- **Windows:** Search for "QGIS Desktop" in the Start menu and click it.
- **Mac:** Open Applications and double-click QGIS.
- **Ubuntu:** Search for "QGIS" in the Activities menu, or type `qgis` in a terminal.

Verify QGIS

Step 1: When QGIS opens, you should see a window with a large blank map area in the centre, a "Layers" panel on the left, and a "Browser" panel.

Step 2: In the menu bar, click **Help** → **About**. Note the version number — it should be 3.34 or newer.

Step 3: Close the About window. QGIS is ready.

> **Tip:** The first time you open QGIS it may take 30–60 seconds to start. This is normal — it is loading plugins and setting up your profile.

Part 3: Test CRD Access

The Climate Risk Database (CRD) is a live web platform that hosts geographic data for disaster resilience in Tanzania. Let us verify you can access it.

Step 1: Open your web browser (Chrome or Firefox).

Step 2: In the address bar, type the following and press Enter:

```
https://crd.resilienceacademy.ac.tz
```

What this does: Opens the live CRD website hosted by Resilience Academy.

You should see: The CRD homepage with a map of Tanzania, a search bar, and categories of datasets listed.

Step 3: Click on **"Data"** (or **"Datasets"**) in the top menu.

You should see: A list of geographic datasets. You can search or filter by category.

Step 4: Try searching for "Mwanza" in the search box.

You should see: Several datasets related to Mwanza appearing in the results.

Step 5: Click on any dataset and explore its details page. Note the download options and the map preview.

> **Good to know:** This is the same platform you will install on your own laptop in Practical 2. The live version at `crd.resilienceacademy.ac.tz` has real data; your local version will start empty and you will learn to upload data to it.

Part 4: Linux Basics Exercise

Whether you use Windows, Mac, or Linux, you will need to type commands in a terminal for this training. This section teaches you the 10 most important commands. Open your terminal now:

- **Windows:** Open **PowerShell** (search in Start menu). Most commands below also work in PowerShell.
- **Mac:** Open **Terminal** (Applications → Utilities → Terminal).
- **Ubuntu:** Press `Ctrl+Alt+T`.

Command 1: pwd — Print Working Directory

Step 1: Type the following and press Enter:

```
pwd
```

What this does: Shows the full path of the folder you are currently in. Think of it as asking "Where am I?"

You should see: Something like `/home/yourname` (Ubuntu/Mac) or `C:\Users\yourname` (Windows).

Command 2: ls — List Files

Step 2: List the files in your current folder:

```
ls
```

What this does: Shows all files and folders in your current location. Like opening a folder in the file manager.

You should see: A list of file and folder names such as `Desktop`, `Documents`, `Downloads`, etc.

Step 3: Now try listing with more detail:

```
ls -la
```

What this does: Shows a detailed list including file sizes, dates, and hidden files (those starting with a dot).

You should see: A table with columns for permissions, owner, size, date, and name.

Command 3: cd — Change Directory

Step 4: Move into the Desktop folder:

```
cd Desktop
```

What this does: Moves you into the Desktop folder, like double-clicking a folder in the file manager.

You should see: No output (silence means it worked). Your prompt may change to show the new location.

Step 5: Confirm your new location:

```
pwd
```

You should see: Something like `/home/yourname/Desktop`.

Step 6: Go back to the previous folder:

```
cd ..
```

What this does: Moves you up one level (to the parent folder). The two dots `..` always mean "parent folder".

You should see: No output. Use `pwd` to confirm you are back.

Command 4: mkdir — Make Directory

Step 7: Create a new folder for this training:

```
mkdir sdi-training
```

What this does: Creates a new empty folder called "sdi-training" in your current location.

You should see: No output (silence means success).

Step 8: Verify the folder was created:

```
ls
```

You should see: The folder `sdi-training` in the list.

Step 9: Move into the new folder:

```
cd sdi-training
```

Command 5: cp — Copy Files

Step 10: First, let us create a small test file (we will learn the tool used here shortly):

```
echo "Hello SDI Training" > testfile.txt
```

What this does: Creates a new file called `testfile.txt` containing the text "Hello SDI Training".

You should see: No output.

Step 11: Copy the file:

```
cp testfile.txt testfile_backup.txt
```

What this does: Makes an exact copy of `testfile.txt` and names the copy `testfile_backup.txt`.

You should see: No output.

Step 12: Verify both files exist:

```
ls
```

You should see: `testfile.txt` and `testfile_backup.txt`.

Command 6: mv — Move or Rename Files

Step 13: Rename a file:

```
mv testfile_backup.txt notes.txt
```

What this does: Renames `testfile_backup.txt` to `notes.txt`. The `mv` command is used both for moving and renaming files.

You should see: No output.

Step 14: Verify the rename:

```
ls
```

You should see: `testfile.txt` and `notes.txt` (the backup name is gone).

Command 7: rm — Remove Files

Step 15: Delete the notes file:

```
rm notes.txt
```

What this does: Permanently deletes `notes.txt`. Unlike the Recycle Bin, this cannot be undone.

You should see: No output.

Step 16: Verify it is gone:

```
ls
```

You should see: Only `testfile.txt` remains.

Warning: The `rm` command permanently deletes files. There is no Recycle Bin or Undo. Always double-check the filename before pressing Enter.

Command 8: cat — Display File Contents

Step 17: View the contents of a file:

```
cat testfile.txt
```

What this does: Prints the entire contents of the file to the screen. Useful for quickly checking what is inside a file.

You should see:

```
Hello SDI Training
```

Command 9: nano — Simple Text Editor

Step 18: Open the file in the nano text editor:

```
nano testfile.txt
```

What this does: Opens a simple text editor inside the terminal. You can type and edit text just like in Notepad.

You should see: The terminal changes to show the file contents with a menu bar at the bottom showing commands like `^X Exit`, `^O Write Out`.

Step 19: Use your arrow keys to move the cursor to the end of the text. Type a new line:

```
My name is [your name]
```

Step 20: Save and exit:

- Press `Ctrl+O` (the letter O, not zero) to save
- Press `Enter` to confirm the filename
- Press `Ctrl+X` to exit nano

Step 21: Verify your edit:

```
cat testfile.txt
```

You should see:

```
Hello SDI Training
My name is [your name]
```

Tip: In nano, the ^ symbol means the `Ctrl` key. So `^X` means press `Ctrl` and `X` together.

Command 10: curl — Fetch Data from the Internet

Step 21: Fetch a web page from the terminal:

```
curl -s https://crd.resilienceacademy.ac.tz/api/v2/ | head -20
```

What this does: `curl` downloads data from a web address, just like a browser but as text. The `-s` flag means "silent" (no progress bar). The `| head -20` part shows only the first 20 lines.

You should see: JSON data (text in curly braces) from the CRD API — something like:

```
{
  "datasets": "https://crd.resilienceacademy.ac.tz/api/v2/datasets/",
  "maps": "https://crd.resilienceacademy.ac.tz/api/v2/maps/",
  ...
}
```

Step 22: Now fetch a list of datasets:

```
curl -s "https://crd.resilienceacademy.ac.tz/api/v2/datasets/?page_size=3" | head
    -50
```

What this does: Requests 3 datasets from the CRD API so you can see what dataset information looks like in its raw form.

You should see: JSON data listing dataset titles, descriptions, and links.

Congratulations! You have just used the CRD API from the command line. APIs (Application

Programming Interfaces) let computer programs talk to each other. The CRD website uses this same API behind the scenes to show you datasets.

Final Checkpoint

Before moving on to Practical 2, confirm all of the following:

- ☐ **Docker:** `docker run hello-world` completes successfully
- ☐ **QGIS:** Opens and shows the main window with a blank map canvas
- ☐ **CRD Access:** `https://crd.resilienceacademy.ac.tz` loads in your browser
- ☐ **Linux Commands:** You practised all 10 commands (`pwd`, `ls`, `cd`, `mkdir`, `cp`, `mv`, `rm`, `cat`, `nano`, `curl`)
- ☐ **API Test:** `curl` returned JSON data from the CRD API

If any item above is not working, raise your hand and the instructor will help you. It is essential that everything works before Day 2 because we will build on these tools.

End of Practical 1 — Proceed to Practical 2: Installing CRD with Docker